04_Capstone_Incident_Report

1. Detection and Triage Log

The following initial alert was generated by the SIEM and triaged by the Tier 1 SOC Analyst, triggering the formal Incident Response process.

Timestamp	Source IP (Attacker)	Target IP (Victim)	Alert Description	MITRE Tactic	Priority
8:20 AM	10.0.2.15	10.0.2.3	VSFTPD v2.3.4 Backdoor Exploit	T1190	Critical

2. Full Incident Report (Technical)

Executive Summary: On March 20, 2026, the Security Operations Center (SOC) detected a critical security incident involving an internal Linux-based file server (IP: 10.0.2.3). A threat actor utilized a known malicious backdoor present in legacy VSFTPD v2.3.4 software to execute a payload, granting them unauthorized root-level access to the system. The endpoint was rapidly isolated, and forensic evidence was preserved for analysis.

Incident Timeline:

- **08:20 AM:** Attacker IP (10.0.2.15) initiates a connection on TCP port 21. Malicious payload deployed triggering the VSFTPD backdoor.
- **08:20 AM:** Command shell session 1 opened. Attacker verifies UID 0 (root) access. SIEM alert generated.
- **08:23 AM:** SOC Analyst identifies the breach, classifies the alert as Critical, and physically/logically isolates the server from the network.
- **08:24 AM:** SOC Analyst captures volatile network connections (ss -antp) to check for persistence or lateral movement.
- **08:29 AM:** Memory dump acquired and cryptographically hashed (SHA256) to establish a formal chain of custody.

Fig. 1: Proof of compromise. The threat actor successfully establishing a reverse shell and verifying root-level privileges on the target file server (10.0.2.3).

Impact Analysis: The threat actor achieved full system compromise (root access) on the target endpoint. However, due to the rapid isolation of the server within three minutes of the initial breach, no lateral movement, privilege escalation to other network segments, or data exfiltration occurred.
Remediation and Eradication Steps: The compromised server (10.0.2.3) has been taken offline and will be entirely rebuilt from a known-good backup image. The deprecated VSFTPD v2.3.4 service will be permanently uninstalled and replaced with a modern, secure file transfer protocol (e.g., SFTP via SSH) with key-based authentication enforced.

Lessons Learned & Preventative Measures: Legacy and deprecated services running on internal networks pose a critical, easily exploitable risk. To prevent future occurrences, the organization must implement a strict, automated weekly vulnerability scanning cadence to identify and patch unsupported software across all environments before exploitation can occur.

3. Stakeholder Briefing (Non-Technical Executive Summary)

At approximately 8:20 AM today, our automated security monitoring tools detected an unauthorized user gaining access to one of our internal file servers. The security operations team acted immediately, isolating the affected server from the rest of the company network within three minutes to completely contain the threat.
Because of this rapid, by-the-book response, we have confirmed that the intruder was unable to access any other parts of our network, and no sensitive company data or customer information was compromised or stolen.

We are currently rebuilding the affected server with updated security protocols. Additionally, we have initiated a comprehensive sweep of all internal systems to ensure any similar outdated software is immediately updated or removed to prevent this from happening again. Operations are continuing as normal with no expected downtime for end-users.